

Saumya Bajaj

Agentic AI & Predictive Anomaly Detection in Healthcare Operations

Healthcare Treatment, Payment, and Operations (TPO) face massive optimization challenges. While traditional payment integrity relies on rigid, non-semantic rules, unconstrained agentic AI systems often result in high costs and hallucination risks. The solution lies in pairing deterministic algorithms with pattern-optimized semantic screening agents to scale fraud, waste, and abuse (FWA) interception.

Modern clinical payment integrity requires an integrated approach to pattern recognition, combining classification, prediction, statistical inference, clustering, and time-series anomaly detection. Rather than evaluating claims as isolated transactions, pattern-optimized sophisticated frameworks apply predictive analytics to identify variance vectors across historic provider benchmarks.

However, mathematical anomaly detection captures only half of the compliance landscape. True validation requires an intelligent semantic layer capable of chain-of-thought reasoning to extract the contextual intent of an encounter. Verifying whether an administrative code accurately mirrors a physician's unstructured clinical chart notes requires bridging the gap between narrative text and alphanumeric standards like ICD-10 and CPT.

While program integrity historically relied on a retrospective "pay-and-chase" model, where claims were audited post-disbursement and recovered through resource-intensive appeals, now detection routines are in the pre-adjudication window to intercept irregular billing behaviors before improper claims are paid (Saxena, 2026). Simultaneously, payers are leveraging AI to control Medical Loss Ratios by capturing subtle discrepancies like modifier misuse or

unbundling (Cognizant Global Research, 2026). Information lag between clinical policy and investigation units further necessitates real-time automated updates (SAS Institute, 2025).

Despite the potential of generative agents, empirical benchmarks show that unconstrained systems offer only modest reasoning improvements while introducing high latency and token overhead (Velasquez et al., 2026). Safety is also a concern, as standalone medical deployment risks "token-propagation hallucinations," where models fail to link disparate data points, such as body-side laterality, within chart notes (Ahmad et al., 2026).

Cotiviti should reject the trend of using a single, heavy LLM for clinical auditing, or even multiple LLM-based agents. Instead, implement an asymmetric, hybrid pipeline. This involves using fast Python math engines to screen 100% of claims for deterministic violations. Claims passing these thresholds clear instantly, while minor variances route to a lightweight semantic worker.

Cotiviti's semantic agents must be decoupled from abstract medical lookup tasks to mitigate latency and cost barriers. Prompt architectures should be engineered around explicit, structural logical conditions and few-shot in-context examples. By instructing the model to cross-reference literal alphanumeric code patterns against explicit keywords in clinical notes, the task becomes a string-matching condition, eliminating the need for expensive parameter weights.

All agent outputs must pass through strict validation gates before hitting downstream databases to address fragmented safety frameworks. By enforcing type-safe Pydantic configurations, Cotiviti can trap open-ended text outputs and convert them into rigid, structured JSON logs. This ensures every automated action item, clinical justification, and risk classification score is explicitly validated against a predefined enterprise schema, completely eliminating model drift and text-based hallucinations.

References

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